ARJUN BHATTACHARYA, PHD

Assistant Professor of Epidemiology · MD Anderson Cancer Center

The Bhattacharya Lab for Computational Genomics

abhattacharya3@mdanderson.org · bhattacharya-lab.com · github.com/bhattacharya-a-bt · ORCID 0000-0003-1196-4385

APPOINTMENTS

Assistant Professor, Department of Epidemiology 2023–
Affiliate, Institute for Data Science in Oncology
UT MD Anderson Cancer Center · Houston, TX

Fellow 2021–23
UCLA Institute for Quantitative & Computational Biosciences

Postdoctoral Fellow 2020–23
Pathology & Laboratory Medicine, UCLA · Los Angeles, CA

EDUCATION

PhD, Biostatistics 2015–20
University of North Carolina at Chapel Hill

BS, Mathematical Decision Sciences · BS, Biology 2011–15
University of North Carolina at Chapel Hill
Mackenzie Family Foundation Innovation Scholar

GRANT FUNDING

ACTIVE

NIH/NIEHS R21 ES039048 PI **$491,915**
Identifying PFAS-induced disruption of placental imprinting and neurodevelopment through multi-omic negative control calibration
09/2026 – 08/2028

NIH/NCI R01 CA317515 Co-I **$2,756,895**
Multi-ancestry study of mammographic density
08/2026 – 07/2031 · 10% effort Yrs 1–3, 20% Yrs 4–5

COMPLETED

07/2024 – 06/2026 **NIH/NCI R21 CA293419** (PI) · *Alternative splicing and isoform expression as mediators for the genetic etiology of breast cancer* · $402,000

04/2025 **MD Anderson CCSG New Faculty Award** (PI) · *Cancer Center Support Grant pilot funding* · $50,000

06/2024 **UT System Rising STARs Award** (PI) · *Start-up support for computational genomics research* · $109,000

AWARDS & HONORS

2025 Emerging Leader, *British Journal of Cancer*

2025– Academic Editor, *PLOS Genetics*

2024 UT Rising STARs Award

2023 UCLA QCB Collaboratory Service Award

2022 Semifinalist, Epstein Postdoctoral Award, ASHG (top 10%)

2022 James V. Neel Award, International Genetic Epidemiology Society

2022 Most Outstanding Oral Presentation, World Congress of Psychiatric Genetics

2021 O'Malley Alumni Award for Publication Excellence in Population Sciences

2021 UCLA Bruins in Genomics Outstanding Mentorship Award

2021–23 ISPG Early Career Investigator Program Award

2021–23 Fellow, UCLA Institute for Quantitative and Computational Biosciences

2021 SER Conference Scholarship

2019–20 Center for Environmental Health and Susceptibility Training Grant

2018–19 Susan G. Komen Graduate Training Fellowship in Breast Cancer Disparities

2017–18 UNC-CH Department of Biostatistics Tuition Award

2011–15 Mackenzie Family Foundation Innovation Scholarship (full, 4 years)

PUBLICATIONS

42 PEER-REVIEWED

A Bhattacharya · trainee · * equal contribution · ★ first/senior author or trainee-led

2026

1 ★ **Quantification method affects replicability of eQTL analysis, colocalization, and TWAS**
ST Head, ST Bresnahan, Y Chang, V Shah, N Cole, WH Wu, **A Bhattacharya**
Nature Genetics · 2026 ↗ doi.org/10.1038/s41588-026-02827-0

2 ★ **Natural variation in transplacental transfer efficiency exposes distinct transcriptional network architectures of PFAS effects on birth weight and gestational age**
ST Bresnahan, HEJ Yong, MG Drelichman, SN Campbell, AE Trapse, GR Romo, CM Cellini, S Lopez, JK Chan, S Chan, ER Elkin, **A Bhattacharya***, JY Huang*
Environmental Health Perspectives · 2026 ↗ doi.org/10.1021/ehp.6c00646

3 **Comparative 10-year performance of mammography artificial intelligence, polygenic, and clinical breast cancer risk models in the Kaiser Permanente Research Bank**
V Arasu, …, ST Head, **A Bhattacharya**, …, W Sieh
Journal of the National Cancer Institute · 2026 ↗ doi.org/10.1093/jnci/djag154

4 ★ **Long-read transcriptome assembly reveals vast isoform diversity in the placenta associated with metabolic and endocrine function**
ST Bresnahan, H Yong, A Nemani, WH Wu, S Lopez, J Kok Yen Chan, F White, P Jacques, M Hivert, S Chan, MI Love, JY Huang*, **A Bhattacharya***
Nature Communications · 2026 ↗ doi.org/10.1038/s41467-026-71303-4

2025

5 ★ **Isoform-level analyses of 6 cancers uncover extensive genetic risk mechanisms undetected at the gene-level**
Y Chang, ST Bresnahan, ST Head, T Harrison, Y Yu, C Huff, B Pasaniuc, S Lindström, **A Bhattacharya**
British Journal of Cancer · 2025 Emerging Leaders in Cancer Special Edition ↗ doi.org/10.1038/s41416-025-03141-y

2024

6 **Splicing-specific transcriptome-wide association uncovers genetic mechanisms for schizophrenia**
J Hervoso, K Amoah, J Dodson, M Choudhury, **A Bhattacharya**, G Quinones-Valdez, B Pasaniuc, G Xiao
American Journal of Human Genetics · 2024 PMID: 38925119 ↗ doi.org/10.1016/j.ajhg.2024.06.001

7 **Enhanced CTLA-4 blockade anti-tumor immunity with APG-157 combination in a murine head and neck cancer**
D Shin, S Basak, M Veena, B Comin-Anduix, **A Bhattacharya**, T Dong, A Ko, P Han, J Jacobs, N Moatamed, L Avila, M Pellegrini, M Wang, E Srivatsan
Cancer Medicine · 2024 PMID: 38686626 ↗ doi.org/10.1002/cam4.7212

8 **Cross-ancestry, cell-type-informed atlas of gene, isoform, and splicing regulation in the developing human brain**
C Wen, M Margolis, R Dai, P Zhang, P Przytycki, D Vo, **A Bhattacharya**, N Matoba, C Jiao, M Kim, E Tsai, C Hoh, N Aygün, R Walker, C Chatzinakos, D Clarke, H Pratt, PsychENCODE Consortium, M Peters, M Gerstein, N Daskalakis, Z Weng, A Jaffe, J Kleinman, T Hyde, D Weinberger, N Bray, N Sestan, D Geschwind, K Roeder, A Gusev, B Pasaniuc, J Stein, M Love, K Pollard, C Liu*, M Gandal*
Science · 2024 PMID: 38781368 ↗ doi.org/10.1126/science.adh0829

9 **Brain cell-type shifts in Alzheimer's disease, autism, and schizophrenia interrogated using methylomics and genetics**
C Yap, D Vo, M Heffel, **A Bhattacharya**, C Wen, Y Yang, K Kemper, J Zeng, Z Zheng, Z Zhu, E Hannon, D Vellame, A Franklin, C Caggiano, B Wamsley, D Geschwind, N Zaitlen, A Gusev, B Pasaniuc, J Mill, C Luo, M Gandal
Science Advances · 2024 PMID: 38781333 ↗ doi.org/10.1126/sciadv.adn7655

10 **Genome-wide association meta-analysis identifies novel ancestry-specific primary open-angle glaucoma loci and shared biology with vascular mechanisms and cell proliferation**
V Lo Faro, **A Bhattacharya**, W Zhou, D Zhou, Y Wang, K Läll, M Kanai, E Lopera-Maya, P Straub, P Pawar, R Tao, X Zhong, S Namba, the Global Biobank Meta-analysis Initiative, S Sanna, IM Nolte, Y Okada, N Ingold, S MacGregor, H Snieder, I Surakka, J Shortt, C Gignoux, N Rafaels, K Crooks, A Verma, SS Verma, L Guare, DJ Rader, C Willer, AR Martin, MA Brantley Jr, ER Gamazon, NM Jansonius, K Joos, NJ Cox, J Hirbo
Cell Reports Medicine · 2024 PMID: 38382466 ↗ doi.org/10.1016/j.xcrm.2024.101430

11 ★ **Polygenic scores for tobacco use provide insights into systemic health risks in a diverse EHR-linked biobank in Los Angeles**
V Venkateswaran, K Boulier, Y Ding, R Johnson, **A Bhattacharya**[*], B Pasaniuc[*]
Translational Psychiatry · 2024 PMID: 38238290 ↗ doi.org/10.1038/s41398-024-02743-z

2023 12 ★ **Isoform-level transcriptome-wide association uncovers extensive novel genetic risk mechanisms for neuropsychiatric disorders in the human brain**
A Bhattacharya, D Vo, C Jops, M Kim, C Wen, J Hervoso, B Pasaniuc[*], M Gandal[*]
Nature Genetics · 2023 PMID: 38036788 ↗ doi.org/10.1038/s41588-023-01560-2

13 **Multi-ancestry genome-wide association study of cannabis use disorder yields insight into disease biology and public health implications**
D Levey, M Galimberti, J Deak, F Wendt, **A Bhattacharya**, D Koller, K Harrington, R Quaden, E Johnson, M Cooke, V Rajagopal, S Empke, H Zhou, Y Nunez, H Kranzler, H Edenberg, A Agrawal, J Smoller, D Demontis, VA Million Veteran Program, J Gaziano, M Gandal, R Polimanti, M Stein, J Gelernter
Nature Genetics · 2023 PMID: 37985822 ↗ doi.org/10.1038/s41588-023-01563-z

14 **Genotype error due to low-coverage sequencing induces uncertainty in polygenic scoring**
E Petter, Y Ding, K Hou, **A Bhattacharya**, A Gusev, N Zaitlen, B Pasaniuc
American Journal of Human Genetics · 2023 PMID: 37490908 ↗ doi.org/10.1016/j.ajhg.2023.06.015

15 **Impact of cross-ancestry genetic architecture on GWAS in admixed populations**
R Mester, K Hou, Y Ding, G Meeks, K Burch, **A Bhattacharya**, B Henn, B Pasaniuc
American Journal of Human Genetics · 2023 PMID: 37224807 ↗ doi.org/10.1016/j.ajhg.2023.05.001

16 **twas_sim, a Python-based tool for simulation and power analysis of transcriptome-wide association analysis**
X Wang, Z Lu, **A Bhattacharya**, B Pasaniuc, N Mancuso
Bioinformatics · 2023 PMID: 37099718 ↗ doi.org/10.1093/bioinformatics/btad288

17 **Causal effects on complex traits are similar across segments of different continental ancestries within admixed individuals**
K Hou, Y Ding, Z Xu, Y Wu, **A Bhattacharya**, R Mester, G Belbin, S Buyske, D Conti, B Darst, M Fornage, C Gignoux, X Guo, C Haiman, E Kenny, M Kim, C Kooperberg, L Lange, A Manichaikul, K North, U Peters, L Rasmussen-Torvik, S Rich, J Rotter, H Wheeler, G Wojcik, Y Zhou, the PAGE Consortium, S Sankararaman, B Pasaniuc
Nature Genetics · 2023 PMID: 36941441 ↗ doi.org/10.1038/s41588-023-01338-6

18 **The UCLA ATLAS Community Health Initiative: promoting precision health research in a diverse biobank**
R Johnson, Y Ding, **A Bhattacharya**, A Chiu, C Lajonchere, D Geschwind, B Pasaniuc
Cell Genomics · 2023 PMID: 36777178 ↗ doi.org/10.1016/j.xgen.2022.100243

2022 19 **Broad transcriptomic dysregulation occurs across the cerebral cortex in ASD**
M Gandal[*], J Haney[*], B Wamsley, C Yap, S Parhami, P Emani, N Chang, G Chen, G Hoftman, D de Alba, G Ramaswami, C Hartl, **A Bhattacharya**, C Luo, T Jin, D Wang, R Kawaguchi, D Quintero, J Ou, Y Wu, N Parikshak, V Swarup, T Belgard, M Gerstein, B Pasaniuc, D Geschwind
Nature · 2022 PMID: 36323788 ↗ doi.org/10.1038/s41586-022-05377-7

20 ★ **Best practices of multi-ancestry, meta-analytic transcriptome-wide association studies: lessons from the Global Biobank Meta-analysis Initiative**
A Bhattacharya[*], J Hirbo[*], D Zhou, W Zhou, J Zheng, M Kanai, the Global Biobank Meta-analysis Initiative, B Pasaniuc, E Gamazon, N Cox
Cell Genomics · 2022 PMID: 36341024 ↗ doi.org/10.1016/j.xgen.2022.100180

21 **Proteome-wide Mendelian randomization in global biobank meta-analysis reveals trans-ancestry drug targets for common diseases**
H Zhao, H Rasheed, T Nost, Y Cho, Y Liu, L Bhatta, **A Bhattacharya**, the Global Biobank Meta-analysis Initiative, G Hemani, G Davey Smith, B Brumpton[*], W Zhou[*], B Neale[*], T Gaunt[*], J Zheng[*]
Cell Genomics · 2022 PMID: 36388766 ↗ doi.org/10.1016/j.xgen.2022.100195

22 **Meta-analysis fine-mapping is often miscalibrated at single-variant resolution**
M Kanai, R Elzur, W Zhou, **the Global Biobank Meta-analysis Initiative**
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100210

23 **Global Biobank analyses provide lessons for developing polygenic risk scores across diverse cohorts**
Y Wang, S Namba, E Lopera, S Kerminen, K Tsuo, K Läll, M Kanai, W Zhou, K Wu, M Favé, L Bhatta, P Awadalla, B Brumpton, P Deelen, K Hveem, V Lo Faro, R Mägi, Y Murakami, S Sanna, J Smoller, J Uzunovic, B Wolford, **the Global Biobank Meta-analysis Initiative**, C Willer, E Gamazon, N Cox, I Surakka, Y Okada, A Martin, J Hirbo
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100241

24 **Leveraging global multi-ancestry meta-analysis in the study of idiopathic pulmonary fibrosis genetics**
J Partanen, P Häppölä, W Zhou, A Lehisto, M Ainola, E Sutinen, R Allen, A Stockwell, O Leavy, J Oldham, B Guillen-Guio, N Cox, J Hirbo, D Schwartz, T Fingerlin, C Flores, I Noth, B Yaspan, R Jenkins, L Wain, S Ripatti, M Pirinen, International IPF Genetics Consortium, **the Global Biobank Meta-analysis Initiative**, T Laitinen, R Kaarteenaho, M Myllärniemi, M Daly, J Koskela
Cell Genomics · 2022 Consortium author ↗ doi.org/10.1016/j.xgen.2022.100181

25 **A practical guideline of genomics-driven drug discovery in the era of global biobank meta-analysis**
S Namba, T Konuma, K Wu, W Zhou, **the Global Biobank Meta-analysis Initiative**, Y Okada
Cell Genomics · 2022 Consortium author · PMID: 36643910 ↗ doi.org/10.1016/j.xgen.2022.100190

26 **Multi-ancestry meta-analysis of asthma identifies novel associations and highlights the value of increased power and diversity**
K Tsuo, W Zhou, Y Wang, M Kanai, S Namba, R Gupta, L Majara, L Nkambule, T Morisaki, Y Okada, B Neale, **the Global Biobank Meta-analysis Initiative**, M Daly, A Martin
Cell Genomics · 2022 Consortium author · PMID: 36778051 ↗ doi.org/10.1016/j.xgen.2022.100212

27 **Global Biobank Meta-analysis Initiative: powering genetic discovery across human diseases**
W Zhou, M Kanai, K Wu, R Humaira, K Tsuo, J Hirbo, Y Wang, **A Bhattacharya**, ..., C Willer, M Daly, B Neale for the Global Biobank Meta-analysis Initiative
Cell Genomics · 2022 PMID: 36777996 ↗ doi.org/10.1016/j.xgen.2022.100192

28 **Leveraging genomic diversity for discovery in an EHR-linked biobank: the UCLA ATLAS Community Health Initiative**
R Johnson, Y Ding, V Venkateswaran, **A Bhattacharya**, K Boulier, A Chiu, S Knyazev, T Schwarz, M Freund, L Zhan, K Burch, C Caggiano, B Hill, N Rakocz, B Balliu, C Denny, J Sul, N Zaitlen, V Arboleda, E Halperin, S Sankararaman, M Butte, UCLA Precision Health Data Discovery Repository Working Group, UCLA Precision Health ATLAS Working Group, C Lajonchere, D Geschwind, B Pasaniuc
Genome Medicine · 2022 PMID: 36085083 ↗ doi.org/10.1186/s13073-022-01106-x

29 **Innovative computational approaches shed light on genetic mechanisms underlying cognitive impairment among children born extremely preterm**
W Liu, Q Sun, L Huang, **A Bhattacharya**, G Wang, X Tan, K Kuban, R Joseph, T O'Shea, R Fry, Y Li, H Santos
Journal of Neurodevelopmental Disorders · 2022 IDDRC Special Issue · PMID: 35240980 ↗ doi.org/10.1186/s11689-022-09429-x

30 ★ **Placental genomics mediates genetic associations with complex health traits and disease**
A Bhattacharya, A Freedman, V Avula, R Harris, W Liu, C Pan, A Lusis, R Joseph, L Smeester, H Hartwell, K Kuban, C Marsit, Y Li, T O'Shea, R Fry, H Santos
Nature Communications · 2022 PMID: 35121757 ↗ doi.org/10.1038/s41467-022-28365-x

31 ★ **Gene-level germline contributions to clinical risk of recurrence scores in Black and White breast cancer patients**
A Patel, M García-Closas, A Olshan, C Perou, M Troester, M Love, **A Bhattacharya**
Cancer Research · 2022 PMID: 34711612 ↗ doi.org/10.1158/0008-5472.CAN-21-1207

32 **Racial differences in breast cancer outcomes by hepatocyte growth factor pathway expression**
G Jones, KA Hoadley, H Benefield, L Olsson, A Hamilton, **A Bhattacharya**, E Kirk, H Tipaldos, J Fleming, K Williams, M Love, H Nichols, A Olshan, M Troester
Breast Cancer Research and Treatment · 2022 PMID: 35034243 ↗ doi.org/10.1007/s10549-021-06497-w

2021

33 **On powerful GWAS in admixed populations**
K Hou, **A Bhattacharya**, R Mester, K Burch, B Pasaniuc
Nature Genetics · 2021 PMID: 34824480 ↗ doi.org/10.1038/s41588-021-00953-5

34 **Hepatocyte growth factor pathway expression in breast cancer by race and subtype**
G Jones, K Hoadley, L Olsson, A Hamilton, **A Bhattacharya**, E Kirk, H Tipaldos, J Fleming, M Love, H Nichols, A Olshan, M Troester
Breast Cancer Research · 2021 PMID: 34344422 ↗ doi.org/10.1186/s13058-021-01460-5

35 ★ **DeCompress: tissue compartment deconvolution of targeted mRNA expression panels using compressed sensing**
A Bhattacharya, A Hamilton, M Troester, M Love
Nucleic Acids Research · 2021 PMID: 33524140 ↗ doi.org/10.1093/nar/gkab031

36 ★ **MOSTWAS: Multi-Omic Strategies for Transcriptome-Wide Association Studies**
A Bhattacharya, Y Li, M Love
PLOS Genetics · 2021 PMID: 33684137 ↗ doi.org/10.1371/journal.pgen.1009398

2020 37 **Biopsychosocial correlates of psychological distress in Latina mothers**
H Santos, H Adynski, R Harris, **A Bhattacharya**, A Incollingo-Rodriguez, R Cali, A Torres Yabar, B Nephew, C Murgatroyd
Journal of Affective Disorders · 2020 PMID: 33445084 ↗ doi.org/10.1016/j.jad.2020.12.193

38 ★ **Evidence for the placenta-brain axis: multi-omic kernel aggregation predicts intellectual and social impairment in children born extremely preterm**
H Santos*, **A Bhattacharya***, R Joseph, L Smeester, K Kuban, C Marsit, T O'Shea, R Fry
Molecular Autism · 2020 PMID: 33308293 ↗ doi.org/10.1186/s13229-020-00402-w

39 ★ **An approach for normalization and quality control for NanoString RNA expression data**
A Bhattacharya*, A Hamilton*, H Furberg, E Pietzak, M Purdue, M Troester, K Hoadley, M Love
Briefings in Bioinformatics · 2020 PMID: 32789507 ↗ doi.org/10.1093/bib/bbaa163

40 ★ **A framework for transcriptome-wide association studies in breast cancer in diverse study populations**
A Bhattacharya, M García-Closas, A Olshan, C Perou, M Troester, M Love
Genome Biology · 2020 PMID: 32079541 · 2021 O'Malley Alumni Award for Publication Excellence ↗ doi.org/10.1186/s13059-020-1942-6

2019 41 **Epigenome-wide DNA methylation in placentas from preterm infants: association with maternal socioeconomic status**
H Santos, **A Bhattacharya**, E Martin, K Addo, M Psioda, L Smeester, R Joseph, S Hooper, J Frazier, K Kuban, T O'Shea, R Fry
Epigenetics · 2019 PMID: 31062658 ↗ doi.org/10.1080/15592294.2019.1614743

2018 42 **Discrimination exposure and DNA methylation of stress-related genes in Latina mothers**
H Santos, B Nephew, **A Bhattacharya**, X Tan, L Smith, R Alyamani, E Martin, K Perreira, R Fry, C Murgatroyd
Psychoneuroendocrinology · 2018 PMID: 30144780 ↗ doi.org/10.1016/j.psyneuen.2018.08.014

PREPRINTS & SUBMITTED 8

2026 1 ★ **ICONIC: An R package for integrating instrumental variable- and negative-control-informed causal discovery and diagnostics in multiomic studies**
ST Bresnahan, C Xiong, ST Head, Y Chang, **A Bhattacharya***, JY Huang*
medRxiv · 2026 ↗ doi.org/10.64898/2026.08.26.26361466

2 ★ **Improving isoform-level eQTL and integrative genetic analyses of breast cancer risk with long-read RNA transcript assemblies**
ST Head, A Nemani, Y Chang, TA Harrison, ST Bresnahan, JH Rothstein, W Sieh, S Lindström, **A Bhattacharya**
bioRxiv · 2026 In review, Nature Communications ↗ doi.org/10.64898/2026.03.22.713514

3 **Cell-type aware transcriptome-wide association study of mammographic density phenotypes**
A Sistig*, JH Rothstein*, S Zhu, ST Head, Y Chang, N Achacoso, SE Alexeeff, V Arasu, T Gadgil, LD Gerstley, LR Margolies, LC Sakoda, L Shen, CLS Gueye, M Villaseñor, M Westley, MODE/BCAC Consortia, **A Bhattacharya**, RJ Klein, LA Habel, X Song, P Wang, W Sieh
medRxiv · 2026 In review, Nature Communications ↗ doi.org/10.1101/2025.11.25.25341027

4 **Integrative proteogenomic analyses reveal insights into subtype-specific glioma risk**
T Nakase, K Smith-Byrne, Q Ostrom, K Suhre, B Melin, M Rentoft, C Ogamba, G Henderson, C Carlo, **A Bhattacharya**, ST Head, G Guerra, M Wrensch, J Wiencke, R Jenkins, J Eckel-Passow, M Bondy, S Francis, L Kachuri
Submitted · 2026 In review, Nature Communications

5 ★ **Formal statistical replication in lung cancer genome-wide association studies**
Y Chang, J Byun, BR Gorman, RJ Hung, JD McKay, CI Amos, S Pyarajan, **A Bhattacharya**[*], R Sun[*]
medRxiv · 2026 In review, HGG Advances ↗ doi.org/10.1101/2025.10.02.25337130

6 ★ **Distal gene regulation mediated by non-coding RNAs contributes to germline risk for breast and prostate cancer**
N Cole, P Lee, T Schwarz, P Zhang, M Freedman, A Gusev, S Lindström, M Gandal, B Pasaniuc, **A Bhattacharya**
medRxiv ↗ doi.org/10.1101/2022.02.08.22270601

2023 7 ★ **Interplay of serum bilirubin and tobacco smoking with lung and head and neck cancers in a diverse, EHR-linked Los Angeles biobank**
V Venkateswaran, E Petter, K Boulier, Y Ding, **A Bhattacharya**[*], B Pasaniuc[*]
medRxiv · 2023 PMID: 37873378 ↗ doi.org/10.1101/2023.09.29.23296364

2022 8 **Multivariate variance components analysis uncovers genetic architecture of brain isoform expression and novel psychiatric disease mechanisms**
M Kim, D Vo, C Jops, C Wen, A Patowary, **A Bhattacharya**, C Yap, H Zhou, M Gandal
medRxiv · 2022 ↗ doi.org/10.1101/2022.10.18.22281204

SELECTED & INVITED TALKS

39

Improved measurement and methods to understand placental mechanisms mediating maternal exposures on fetal outcomes
Baylor OB/GYN Research Roundtable — Jan 2027
INVITED

Opportunities and advancements in integrating long- and short-read RNA-seq across differential expression analysis, QTL mapping, and outcome prediction
Joint Statistical Meetings, Boston — Aug 2026
INVITED

Splicing-resolved genomics for precision women's health across populations
Lunenfeld-Tanenbaum Research Institute, University of Toronto — Jan 2026
INVITED

Discovering novel isoform regulation in complex diseases by integrating long- and short-read RNA-sequencing
Institute for Molecular Medicine Finland (FIMM), HiLIFE, University of Helsinki — Oct 2025
INVITED

Isoform-specific analyses to uncover hidden cancer risk mechanisms
Novel Statistical Approaches for Studying Multi-omics Data, Banff International Research Station — Jul 2025
INVITED

Pacific Center for Genome Research, University of Hawaii Cancer Center — Mar 2025
INVITED

UNC Biostatistics 75th Anniversary Celebration — Nov 2024
INVITED

Vanderbilt University Division of Epidemiology — Sep 2024
INVITED

Integrating environment, gene expression, & computational models to identify modifiable pathways across diverse populations
Pediatric Academic Societies Annual Meeting — Apr 2025
STATE-OF-THE-ART PLENARY

Considerations for computational molecular precision medicine
MD Anderson Department of General Internal Medicine — Jul 2024
INVITED

MD Anderson Division of Cancer Prevention and Control Grand Rounds — Jun 2024
INVITED

Integrative computational methods to unlock the wealth of information in bulk RNA-sequencing
American Statistical Association STATGEN 2024 — May 2024

UNC-Chapel Hill BIOS Genomics Seminar — Jan 2024
INVITED

UTHealth Houston SPH Human Genetics Center — Jan 2024
INVITED

Isoform-level TWAS uncover novel biological mechanisms underlying genetic associations with neuropsychiatric traits
American College of Neuropsychopharmacology Annual Meeting — Dec 2023
INVITED PANEL

UPenn Symposium, Genetics and Epigenetics of Behavior — May 2023
INVITED

ASHG Annual Meeting — Oct 2022
PLATFORM · EPSTEIN AWARD SEMIFINALIST

Harvard Program in Genetic Epidemiology and Statistical Genetics — Oct 2022
INVITED

World Congress of Psychiatric Genetics — Sep 2022
ORAL · MOST OUTSTANDING ORAL PRESENTATION

International Genetic Epidemiology Society — Sep 2022
ORAL · JAMES V. NEEL AWARD

Gene-level germline associations with breast cancer subtype and survival

Breast Ovarian Consortia Meeting — Jun 2023
INVITED

Leveraging the breadth and depth of transcriptomics to understand the genetic etiology of complex traits

NYU Center of Human Genetics and Genomics — Feb 2023
INVITED

MD Anderson Department of Cancer Epidemiology — Jan 2023
INVITED

Emory School of Medicine Human Genetics — Jan 2023
INVITED

UNC Computational Medicine — Jan 2023
INVITED

USC Keck Center of Genetic Epidemiology — Dec 2022
INVITED

Carnegie Mellon Computational Biology — Dec 2022
INVITED

Lieber Institute for Brain Development — Nov 2022
INVITED

Regeneron Genetics Center, Statistical Genetics — Oct 2022
INVITED

Distal mediator-enriched, placental TWAS of 40 traits suggest genetic mechanisms supporting the DOHaD hypothesis

Vanderbilt University Medical Center Division of Epidemiology — Dec 2022
INVITED

US DOHaD Society Annual Meeting — Nov 2021
JUNIOR INVESTIGATOR KEYNOTE

ASHG Annual Meeting — Oct 2021
PLATFORM

World Congress of Psychiatric Genetics — Oct 2021
ORAL · ECIP

Singapore Institute for Clinical Sciences — Aug 2021
INVITED

Society for Epidemiologic Research — Jun 2021
ORAL

UCLA QCBio Research Seminar — Apr 2021
INVITED

DeCompress: tissue compartment deconvolution for targeted RNA panels using compressed sensing

ICCABS, Computational Advances for NGS Workshop — Dec 2020
INVITED

MOSTWAS: Multi-Omic Strategies for Transcriptome-Wide Association Studies

Society for Epidemiologic Research — Dec 2020
ORAL

ASHG Annual Meeting — Oct 2020
PLATFORM

MENTORSHIP

POSTDOCTORAL SCHOLARS

Taylor Head, PhD
Current

Sean Bresnahan, PhD
Current

GRADUATE STUDENTS

Yung-Han Chang
PhD, Quantitative Sciences, MDA GSBS · co-advised with Ryan Sun

Yan Zhang
PhD, Quantitative Sciences, MDA GSBS · co-advised with Callie Kwartler · AHA Predoctoral Fellow

Jacob Smith
CPRTP Summer Program

Achal Patel
PhD, Epidemiology, UNC → Data Scientist, Genentech

Vidhya Venkateswaran
PhD, Oral Biology, UCLA → Senior Research Fellow & NIH DATA Scholar, NIDCR

UNDERGRADUATES

Aryun Nemani
Rice University

Charis Xiong
Rice University

Vihan Shah
UCSD

William Wu
Rice → MD, UT Southwestern (full scholarship)

Erika Vasquez
Rice → Research Associate, NIDCR

Nolan Cole
UCLA BIG → Biostatistics PhD, U. Washington

Paige Lee
UCLA BIG → MS Data Science, Harvard

SERVICE & TEACHING

EDITORIAL & REVIEW

Academic Editor, *PLOS Genetics*

Grant review: NIH NIDA ZDA1 DRO-N (Member); Swiss National Science Foundation

Referee: Nature, Nature Genetics, Nature Biotechnology, Nature Communications, AJHG, eLife, PLOS Genetics

Abstract review: SER Annual Meetings; ASHG 2023

Reviewer, MD Anderson NCI Summer Research Trainee Program, 2024

ACADEMIC SERVICE

MDA GSBS Student Success & Well-being Committee — Member 2024–, **Chair 2026–**

TEACHING

UCLA QCB W5, RNA-seq Analysis · 2021–2023

BIOS 735, Introduction to Data Science · 2019

BIOS 550, Basic Elements of Probability & Statistical Inference · 2018

BIOS 673, Probability & Statistics · 2017

COMPUTING

Advanced: R (incl. package development), SAS, LaTeX, HPC

Intermediate: Python, C++, MATLAB

OPEN-SOURCE SOFTWARE

`isotwas`
Isoform-level transcriptome-wide association studies

`MOSTWAS`
Distal-mediator enriched TWAS

`DeCompress`
Cell-type deconvolution for targeted RNA panels

`NanoNormIter`
QC & normalization for NanoString nCounter